VCTERL Syndrome (WBP11-Related): Comprehensive Disease Characteristics Report

Disease: Vertebral, Cardiac, Tracheoesophageal, Renal, and Limb Defects (VCTERL) **MONDO:** MONDO:0030987 · **OMIM:** 619227 · **Causal gene: WBP11** (HGNC:16461) **Inheritance:** Autosomal dominant (haploinsufficiency)

Summary

VCTERL syndrome is an autosomal-dominant multiple-congenital-anomaly disorder caused by heterozygous loss-of-function (LoF) or damaging missense variants in *WBP11* (WW domain binding protein 11; HGNC:16461), the gene that encodes a nuclear spliceosome component (also known as SIPP1/NpwBP/Npw38-binding protein) that activates pre-mRNA splicing. The disorder was defined by Martin and colleagues in 2020, who identified *WBP11* variants in seven unrelated families presenting with overlapping cardiac, vertebral, tracheo-esophageal, renal, and limb defects — i.e., the VACTERL malformation spectrum (PMID: 33276377). VCTERL (OMIM:619227) represents the **Mendelian, WBP11-driven subset** of the broader VACTERL association, which is itself predominantly sporadic and multifactorial.

The disease mechanism is **haploinsufficiency**: a single functional copy of *WBP11* is insufficient for normal organogenesis. This is strongly supported by population-genetics constraint data — *WBP11* has a gnomAD probability-of-LoF-intolerance (pLI) of 1.0 (LOEUF 0.29), placing it among the most LoF-intolerant genes in the genome — and by a *Wbp11* mouse model in which homozygous nulls die before embryonic day 8.5 (an essential gene), while heterozygous nulls are small and recapitulate human defects of the axial skeleton, kidneys, and esophagus. A zebrafish *wbp11* loss-of-function model likewise produces multi-system developmental defects modeling VACTERL. As a spliceosome-associated factor, reduced WBP11 dosage is inferred to impair pre-mRNA splicing of developmental regulators during the first-trimester critical window of mesoderm, foregut, and limb-bud patterning.

Because VCTERL is defined by structural malformations present at birth, there is **no disease-specific pharmacotherapy or gene therapy**; management is organ-directed surgical correction and lifelong multidisciplinary supportive care. Prognosis is driven chiefly by the severity of the component malformations — especially cardiac defects — with high survival after successful repair but substantial long-term morbidity, particularly from the esophageal component (dysphagia, reflux, dysmotility, stricture, Barrett's esophagus, respiratory disease, and reduced quality of life). Inheritance shows incomplete penetrance and highly variable expressivity: the same *WBP11* LoF variant can cause full VACTERL-spectrum disease or isolated Klippel-Feil syndrome, renal agenesis, or esophageal atresia.

Key Findings

Finding 1 — WBP11 haploinsufficiency causes VCTERL syndrome

Martin et al. (2020) reported **heterozygous predicted loss-of-function and predicted damaging missense variants in *WBP11* in seven unrelated families** with a variety of overlapping congenital malformations spanning the cardiac, vertebral, tracheo-esophageal, renal, and limb systems — the VCTERL/VACTERL spectrum. This established *WBP11* as the causal gene for OMIM:619227, with autosomal-dominant inheritance operating through haploinsufficiency.

> *"we report novel heterozygous predicted loss-of-function (LoF) and predicted damaging missense variants in the WW domain binding protein 11 (WBP11) gene in seven unrelated families with a variety of overlapping congenital malformations, including cardiac, vertebral, tracheo-esophageal, renal and limb defects"* — PMID: 33276377

This is the defining case series and the primary evidence basis for the entire gene–disease relationship.

Finding 2 — WBP11 encodes a spliceosome component that activates pre-mRNA splicing

WBP11 (aliases **SIPP1, NpwBP, Npw38-binding protein**) encodes a nuclear-speckle-localized component of the spliceosome. Llorian et al. (2004) characterized SIPP1 (= WBP11) as a **pre-mRNA splicing factor and interactor of protein phosphatase-1 (PP1)** via an RVXF docking motif; it binds PQBP-1/Npw38, localizes to nuclear speckles (storage and assembly sites for splicing factors), and a SIPP1 fragment inhibits splicing catalysis in nuclear extracts. Martin et al. (2020) independently describe WBP11 as a splicing-activating spliceosome component. This makes VCTERL a **spliceosomopathy** — a developmental disorder caused by impaired function of the general pre-mRNA splicing machinery.

> *"SIPP1 for splicing factor that interacts with PQBP-1 and PP1"* — PMID: 14640981
>
> *"we found that SIPP1 is also a component of the spliceosomes and that a SIPP1-fragment inhibits splicing catalysis by nuclear extracts"* — PMID: 14640981
>
> *"WBP11 encodes a component of the spliceosome with the ability to activate pre-messenger RNA splicing"* — PMID: 33276377

Finding 3 — A *Wbp11* mouse model recapitulates VCTERL and shows the gene is essential for development

Martin et al. (2020) generated a *Wbp11* null allele by CRISPR-Cas9. **Homozygous null embryos die before E8.5**, confirming *Wbp11* is an essential gene. **Heterozygous null mice are under-represented (embryonic/postnatal lethality), are small, and exhibit defects of the axial skeleton, kidneys, and esophagus** — directly mirroring the affected humans and providing experimental support for haploinsufficiency as the disease mechanism.

> *"Wbp11 homozygous null embryos die prior to E8.5, indicating that Wbp11 is essential for development"* — PMID: 33276377
>
> *"Wbp11 heterozygous null mice are small and exhibit defects in axial skeleton, kidneys and esophagus, similar to the affected individuals"* — PMID: 33276377

A complementary **zebrafish *wbp11* loss-of-function model** (*Loss of wbp11 causes multi-system developmental defects: a zebrafish model of VACTERL association*, PMID: 42625817) was also established, broadening the cross-species evidence that *WBP11* dosage governs multi-organ morphogenesis.

Finding 4 — WBP11 is extremely loss-of-function intolerant (pLI = 1.0), supporting haploinsufficiency

gnomAD constraint metrics for *WBP11* (ENSG00000084463, HGNC:16461, chr12:14,784,582–14,803,486 GRCh38, cytoband 12p12.3–p13.1) place it among the most constrained genes genome-wide:

Metric	Value	Interpretation
pLI	1.00 (0.99999998)	Extreme intolerance to heterozygous LoF
LOEUF (oe_lof_upper)	0.291	Strongly constrained (lower = more constrained)
Observed/expected pLoF	0.18 (12 obs vs 66.7 exp)	~82% depletion of LoF variants
LOF Z	5.68	Highly significant LoF depletion
Missense Z	3.93 (oe_mis = 0.68)	Additional missense constraint

A pLI of 1.0 means essentially no healthy individuals in population databases tolerate a *WBP11* LoF allele without consequence — precisely the signature expected for a haploinsufficient developmental gene, and fully concordant with the human and mouse genetics.

Finding 5 — VCTERL (WBP11) is a Mendelian cause within the broader VACTERL association

The **VACTERL association** is defined clinically by the co-occurrence of **≥3** component features: **V**ertebral anomalies, **A**nal atresia, **C**ardiac malformations, **T**racheo-**E**sophageal fistula ± esophageal atresia, **R**enal anomalies, and **L**imb abnormalities (Solomon et al. 2012). Most VACTERL is **sporadic and multifactorial**, with low familial recurrence (~1%); only ~11–12% of tracheo-esophageal/VACTERL cases have an identifiable genetic cause (Brosens et al. 2014). *WBP11* LoF defines an autosomal-dominant Mendelian subset (VCTERL, OMIM:619227) that

can present as the full VACTERL spectrum **or** as an isolated component phenotype — Klippel-Feil syndrome, renal agenesis, or esophageal atresia. Other established Mendelian VATER/VACTERL genes include *FGF8*, *FOXF1*, *HOXD13*, *LPP*, *TRAP1*, and *ZIC3* (Reutter et al. 2016).

> *"VACTERL association ... is typically defined by the presence of at least three of the following congenital malformations"* — PMID: 23165726
>
> *"At present, a causal genetic aberration can be identified in 11-12% of patients"* — PMID: 24931924
>
> *"LoF WBP11 variants should be considered as a possible cause of VACTERL association as well as isolated Klippel-Feil syndrome, renal agenesis or esophageal atresia"* — PMID: 33276377

Finding 6 — Prognosis is driven by component-malformation severity, especially cardiac defects

Outcome in VCTERL/VACTERL is dominated by the severity of the individual malformations. In esophageal-atresia (EA) cohorts, mortality is substantially higher when **major congenital heart disease (CHD)** is present — 60% mortality among neonates with major CHD and 40% among those with VACTERL association (Varga et al. 2026) — and EA patients with VACTERL plus chromosomal anomalies form a high-risk subgroup marked by prematurity, major cardiac defects, and increased sepsis risk (Soyer et al. 2026). With successful surgical repair, overall survival is now high, but **long-term morbidity is substantial**: adults born with esophageal atresia suffer dysphagia, gastro-esophageal reflux, esophageal dysmotility, anastomotic stricture, Barrett's esophagus (with associated esophageal cancer risk), chronic respiratory disease, feeding/nutritional problems, and reduced quality of life, requiring lifelong follow-up.

> *"Mortality was 60% among neonates with major congenital heart defects (CHD) and 40% among those with VACTERL association"* — PMID: 42463002
>
> *"Many of the adults born with œsophageal atresia will suffer from dysphagia, gastro-œsophageal reflux and/or œsophageal dysmotility, leading to nutritional consequences and quality of life impairment"* — PMID: 35331402
>
> *"EA patients with VACTERL-CA represent a high-risk subgroup characterized by prematurity, major cardiac defects, and increased sepsis risk"* — PMID: 40992427

Finding 7 — Etiology/environment: broad VACTERL is multifactorial; WBP11-VCTERL is monogenic

Non-syndromic VACTERL is generally considered multifactorial, with contributions from epigenetic and environmental factors; most cases are sporadic (~1% familial recurrence) (Brosens et al. 2014). **Maternal pregestational diabetes** is an established risk factor for VACTERL-type anomalies — e.g., it significantly increased the risk of hearing loss in VACTERL (OR 3.71, 95% CI 1.5–7.3) and in partial VACTERL (OR 6.7, 95% CI 1.70–26) (Galarreta et al. 2023). Analyses of recurrent constellations of embryonic malformations note an excess of monozygotic twins discordant for the phenotype and elevated assisted-reproductive-technology (ART) use (~18.5%), implicating early post-zygotic/epigenetic disruption (Adam et al. 2020). In contrast, **WBP11-related VCTERL is monogenic**: no environmental trigger is required, although the same developmental pathways may be modulated by such factors.

> *"non-syndromic EA/TEF is generally believed to be a multifactorial condition"* — PMID: 24931924
>
> *"Maternal diabetes significantly increased the risk for HL in VACTERL (odds ratio [OR]: 3.71, 95% confidence interval [CI]: 1.5-7.3)"* — PMID: 37649433
>
> *"Statistically significant associations included an excess of monozygotic twins and 80% discordance rate for the phenotype across all twins. There was an 18.5% rate of ART"* — PMID: 32924308

Finding 8 — Proposed mechanism: reduced splicing capacity disrupts developmental gene programs during organogenesis

The inferred causal chain (partly demonstrated, partly inferred) runs from the molecular lesion to the clinical phenotype. WBP11 is proposed to intersect VACTERL developmental pathways; a 2025 review places *WBP11* among genes influencing VACTERL alongside Sonic Hedgehog (SHH)-pathway components (*SHH, GLI, ZIC3, FOXF1, HOXD13, FGF8*). The mouse *Wbp11*-null lethality before E8.5 confirms an essential early-developmental role.

> *"WBP11 encodes a component of the spliceosome with the ability to activate pre-messenger RNA splicing"* — PMID: 33276377

> *"we describe the influence of TRAP1, COL11A2, SALL4, WBP11, Copy Number Variants, and maternal environmental factors on VACTERL"* — PMID: 40692799

Finding 9 — ClinVar documents an established WBP11 gene–disease relationship

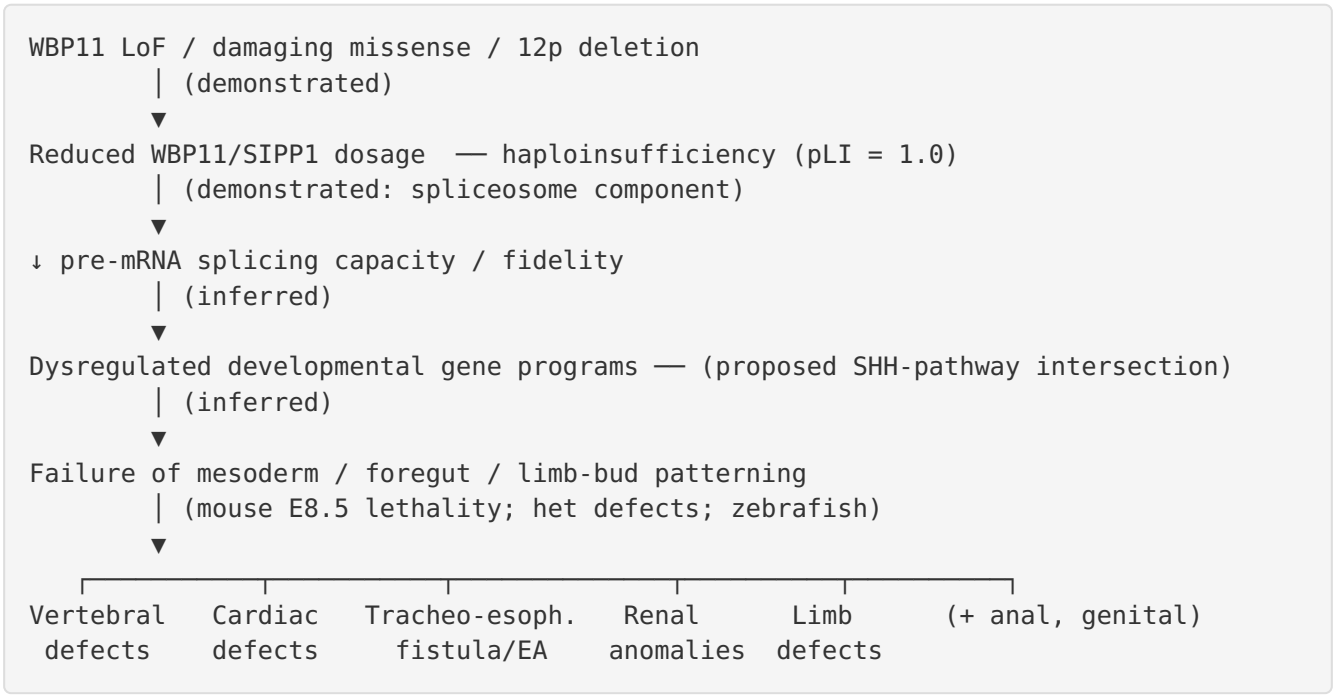
NCBI ClinVar (accessed 2026-10) returns ~228 *WBP11* variant records, of which **100 are classified Pathogenic and 18 Likely Pathogenic**. This corroborates *WBP11* as a well-established, actively curated disease gene for VCTERL/VACTERL-spectrum congenital anomalies, complementing the gnomAD constraint and the defining case series. Reported disease alleles are germline LoF (nonsense/frameshift/splice-site) and damaging missense, plus 12p deletions/CNVs encompassing *WBP11*.

Mechanistic Model / Interpretation

Ordered causal chain (initiating lesion → clinical manifestation)

1. A **heterozygous *WBP11* LoF (nonsense/frameshift/splice) or damaging missense variant** (or a 12p12.3–p13.1 deletion encompassing *WBP11*) arises in the germline. *(demonstrated — PMID: 33276377, ClinVar)*
2. This **leads to** reduced dosage of the WBP11/SIPP1 spliceosome co-factor — i.e., haploinsufficiency. *(demonstrated: gnomAD pLI = 1.0; SIPP1 is a bona-fide spliceosome component — PMID: 14640981)*
3. Reduced WBP11 **results in** globally reduced or altered pre-mRNA splicing efficiency / fidelity. *(inferred — splicing-activating function demonstrated in vitro; a genome-wide splicing defect in affected tissue has not been directly measured)*
4. Altered splicing **impairs** the expression of splicing-sensitive developmental regulators and the proliferation of progenitor cells during the first-trimester critical window of organogenesis. *(inferred; intersection with SHH-pathway genes proposed — PMID: 40692799)*
5. This **causes** failure of mesoderm, foregut, and limb-bud patterning. *(inferred, supported by mouse/zebrafish phenotypes)*

6. The patterning failures **manifest as** vertebral, cardiac, tracheo-esophageal, renal, and limb malformations — the VACTERL/VCTERL spectrum. *(demonstrated in humans and recapitulated in heterozygous Wbp11 mice and wbp11 zebrafish)*



Where the mechanism branches

The disorder's hallmark **variable expressivity and incomplete penetrance** sit at steps 4–6: the same reduction in splicing capacity can disrupt one organ field or several, producing anything from isolated Klippel-Feil syndrome, isolated renal agenesis, or isolated esophageal atresia up to the full ≥3-feature VACTERL phenotype (PMID: 33276377). The branch point is stochastic/modifier-dependent rather than deterministic.

Ontology annotations

Category	Suggested terms
Disease	MONDO:0030987; OMIM:619227
Gene/protein	HGNC:16461 (*WBP11*); UniProt Q9Y2W2; Ensembl ENSG00000084463
Biological process (GO)	GO:0000398 (mRNA splicing, via spliceosome); GO:0008380 (RNA splicing); GO:0048704 (embryonic skeletal system morphogenesis); GO:0072001 (renal system development)
Cellular component (GO)	GO:0005681 (spliceosomal complex); GO:0016607 (nuclear speck); GO:0005634 (nucleus)
Molecular function (GO)	GO:0003723 (RNA binding); GO:0019903 (protein phosphatase binding)
Anatomy (UBERON)	UBERON:0001130 (vertebral column); UBERON:0000948 (heart); UBERON:0003126 (trachea); UBERON:0001043 (esophagus); UBERON:0002113 (kidney); UBERON:0002101 (limb)
Phenotype (HPO)	HP:0003468 (vertebral segmentation defect); HP:0001627 (abnormal heart morphology); HP:0002575 (tracheoesophageal fistula); HP:0002040 (esophageal atresia); HP:0000104 (renal agenesis); HP:0002817 (abnormality of the upper limb); HP:0002023 (anal atresia)
Chemical (CHEBI)	Not applicable (no causative/therapeutic chemical entity)

Section-by-Section Report

1. Disease Information

VCTERL (Vertebral, Cardiac, Tracheo-esophageal, Renal, and Limb Defects) is an autosomal-dominant multiple-congenital-anomaly syndrome caused by heterozygous *WBP11* variants. **Identifiers:** MONDO:0030987; OMIM:619227; synonyms VCTERL / VCTERL syndrome. It is the Mendelian, *WBP11*-defined subset of the broader VATER/VACTERL association (OMIM #192350). No single ICD-10/ICD-11 or MeSH code is specific to VCTERL; cases are coded by their

component malformations or under the VATER-association descriptor (e.g., ICD-10 Q87.2). Information is derived from **aggregated disease-level resources** (case series, OMIM, ClinVar, gnomAD) rather than EHR-level individual records.

2. Etiology

Primary cause: monogenic — heterozygous *WBP11* LoF or damaging missense variants (haploinsufficiency). **Genetic risk factors:** the causal *WBP11* allele itself; other VATER/VACTERL Mendelian genes (*FGF8*, *FOXF1*, *HOXD13*, *LPP*, *TRAP1*, *ZIC3*) cause phenocopies. **Environmental risk factors** (relevant to the broader VACTERL phenotype, not required for WBP11-VCTERL): maternal pregestational diabetes (OR up to ~6.7 for some features), assisted reproduction/twinning. **Protective factors:** none specifically identified; standard periconceptional care and maternal glycemic control reduce general congenital-anomaly risk. **Gene–environment interactions:** proposed modulation of shared developmental pathways (e.g., SHH signaling) by the maternal metabolic environment, but not demonstrated for *WBP11* specifically.

3. Phenotypes

Core phenotypes are structural congenital malformations present at birth (**congenital onset; variable, often severe; stable structurally once repaired**). Frequencies below reflect broader VACTERL ascertainment and *WBP11* case reports.

Phenotype	HPO term	Type	Notes
Vertebral anomalies (incl. Klippel-Feil)	HP:0003468 / HP:0003027	Physical/ radiologic	Can be isolated in WBP11
Cardiac malformation	HP:0001627	Physical sign	Dominant prognostic driver
Tracheo-esophageal fistula ± esophageal atresia	HP:0002575 / HP:0002040	Physical sign	Major long-term morbidity
Renal anomalies (incl. agenesis)	HP:0000104 / HP:0000077	Physical/ imaging	Most common VACTERL CF overall
Limb defects (radial/thumb)	HP:0002817 / HP:0009777	Physical sign	
Anal atresia	HP:0002023	Physical sign	Variable
Genital anomalies	HP:0000078	Physical sign	~34% in VACTERL cohorts

Quality-of-life impact is dominated by the esophageal/GI component (dysphagia, reflux, feeding difficulty) and by cardiac/renal severity; adults report measurable reductions on GIQLI/GSRS instruments.

4. Genetic/Molecular Information

Causal gene: *WBP11* (HGNC:16461; chr12p12.3–p13.1; ENSG00000084463; UniProt Q9Y2W2). **Variant classes:** germline nonsense, frameshift, canonical splice-site, and predicted-damaging missense variants; also 12p deletions/CNVs encompassing *WBP11*. **Classification:** ClinVar lists ~100 Pathogenic and 18 Likely Pathogenic records (ACMG/AMP). **Allele frequency:** extreme LoF depletion in gnomAD (12 observed vs 66.7 expected pLoF; pLI = 1.0; LOEUF 0.29). **Origin:** germline (de novo or inherited). **Functional consequence:** loss of function → haploinsufficiency. **Modifier genes/epigenetics:** not specifically characterized for *WBP11*; variable expressivity suggests unidentified modifiers/stochastic effects.

5. Environmental Information

No environmental agent is required to cause WBP11-VCTERL (monogenic). For the overlapping multifactorial VACTERL phenotype, maternal pregestational diabetes is the best-established environmental/metabolic risk factor; ART/twinning associations implicate early post-zygotic events. No infectious agent is implicated.

6. Mechanism / Pathophysiology

See the **Mechanistic Model** above for the ordered causal chain. In brief: *WBP11* LoF → reduced WBP11/SIPP1 spliceosome co-factor dosage → impaired pre-mRNA splicing (spliceosomopathy) → dysregulated developmental gene programs (proposed SHH-pathway intersection) → failed mesoderm/foregut/limb-bud patterning → multi-organ malformation. **Cellular processes:** progenitor proliferation/patterning during organogenesis. **Protein dysfunction:** loss of a nuclear-speckle splicing-activating factor; PP1 and PQBP-1/Npw38 interactions. **Upstream vs downstream:** the splicing deficit is upstream; organ-specific morphogenetic failure is downstream.

7. Anatomical Structures Affected

Primary organs: vertebral column/axial skeleton (UBERON:0001130), heart (UBERON:0000948), trachea (UBERON:0003126) and esophagus (UBERON:0001043), kidney (UBERON:0002113), limbs (UBERON:0002101); also anus/hindgut and genitalia. **Body systems:** skeletal, cardiovascular, respiratory, digestive, genitourinary. **Subcellular:** nucleus / nuclear speckle / spliceosomal complex (GO:0016607, GO:0005681). **Lateralization:** malformations may be unilateral, bilateral, or asymmetric depending on organ.

8. Temporal Development

Onset: congenital (prenatal organogenesis, first-trimester critical window). **Detection:** prenatal ultrasound detects renal (~45%), TEF (~44%), cardiac (~20%), vertebral (~13%), limb (~11%) features; anal atresia is typically not seen prenatally. **Course:** structural defects are static once repaired, but the esophageal component produces chronic, lifelong GI/respiratory morbidity. **Critical period for intervention:** neonatal surgical correction of life-threatening anomalies (EA/TEF, cardiac, anal atresia).

9. Inheritance and Population

Inheritance: autosomal dominant, haploinsufficiency; de novo or inherited. **Penetrance:** incomplete. **Expressivity:** highly variable (isolated single-organ to full VACTERL spectrum). **Epidemiology:** VCTERL itself is ultra-rare (seven founding families; a few dozen ClinVar families); the broader VACTERL association occurs in ~1 in 10,000–40,000 births, of which only ~11–12% have an identifiable genetic cause, and *WBP11* is one minority Mendelian contributor. No founder effect, anticipation, or strong sex bias is established for *WBP11*.

10. Diagnostics

Clinical: recognition of ≥3 VACTERL component features; imaging (echocardiography, renal ultrasound, spinal/limb radiographs, contrast studies for EA/TEF). **Genetic testing:** exome/genome sequencing or multigene congenital-anomaly panels including *WBP11*; chromosomal microarray to detect 12p deletions/CNVs; single-gene *WBP11* testing when the phenotype is suggestive. **Differential diagnosis:** VACTERL of other genetic cause (*FGF8, FOXF1, HOXD13, LPP, TRAP1, ZIC3*), Fanconi anemia (VACTERL-H — screen when radial/renal defects co-occur; worse prognosis, cancer risk), CHARGE syndrome, Feingold syndrome, trisomy 18/21.

11. Outcome/Prognosis

Survival is now high after surgical repair but is dominated by **cardiac-defect severity** (up to 60% neonatal mortality with major CHD in EA cohorts; VACTERL + chromosomal anomalies = high-risk subgroup). **Long-term morbidity** (esophageal component): dysphagia, GERD, dysmotility, anastomotic stricture, Barrett's esophagus/cancer risk, chronic respiratory disease, feeding/growth problems, and reduced quality of life. Lifelong multidisciplinary surveillance is mandated.

12. Treatment

There is **no disease-specific or gene-directed therapy**. Management is **organ-directed surgical correction** (neonatal repair of EA/TEF, cardiac surgery, anorectal/limb/renal procedures) plus **supportive and rehabilitative care** (nutrition/feeding therapy, anti-reflux management, pulmonary care, physical/occupational/speech therapy). Treatment is individualized by component-malformation profile. NCIT suggestions apply to specific interventions (surgical repair, supportive care) rather than a syndrome-specific agent.

13. Prevention

Primary prevention: not possible for a de novo monogenic disorder; for the broader phenotype, maternal glycemic control and periconceptional care reduce risk. **Secondary prevention:** prenatal ultrasound detection enables delivery planning at a surgical center. **Genetic counseling:** autosomal-dominant recurrence counseling (50% transmission risk from an affected heterozygous parent, with incomplete penetrance/variable expressivity); prenatal/preimplantation genetic testing possible once a familial *WBP11* variant is known. **Cascade testing** of at-risk relatives is appropriate.

14. Other Species / Natural Disease

No naturally occurring companion-animal or wildlife VCTERL disease is documented. The gene is evolutionarily conserved; mouse *Wbp11* (essential; E8.5 lethal when null) and zebrafish *wbp11* are orthologs used experimentally. No zoonotic/transmission relevance.

15. Model Organisms

Mouse: CRISPR-Cas9 *Wbp11* null — homozygous lethal before E8.5; heterozygotes small with axial-skeleton, kidney, and esophageal defects (strong recapitulation of the human disorder) (PMID: 33276377). **Zebrafish:** *wbp11* loss-of-function model with multi-system developmental defects modeling VACTERL (PMID: 42625817). **In vitro:** biochemical characterization of SIPP1/WBP11 as a spliceosome component and PP1/PQBP-1 interactor (PMID: 14640981). **Limitations:** models capture structural malformation but not the full human variable expressivity; genome-wide splicing consequences in affected human tissues remain to be mapped.

Evidence Base

PMID	Title (abbrev.)	Evidence type	Role in report
33276377	*Heterozygous loss of WBP11 function causes multiple congenital defects in humans and mice*	Human clinical + mouse	**Defining paper**: gene–disease relationship, variants, mouse model, splicing function, expressivity
14640981	*SIPP1, a novel pre-mRNA splicing factor and interactor of PP1*	In vitro/ biochemical	Establishes WBP11/SIPP1 molecular function (spliceosome, PP1, PQBP-1)
42625817	*Loss of wbp11 ... zebrafish model of VACTERL*	Model organism	Cross-species confirmation of multi-system phenotype
40692799	*Molecular mechanism, diagnosis, and treatment of VACTERL*	Review	Places WBP11 in VACTERL developmental-pathway context
23165726	*Clinical geneticists' views of VACTERL/VATER*	Clinical consensus	Diagnostic definition (≥3 features)
24931924	*Clinical and etiological heterogeneity in tracheo-esophageal malformations*	Review/cohort	Multifactorial etiology; ~11–12% genetic yield
26857713	*Underlying genetic factors of VATER/VACTERL ... "Renal" phenotype*	Review	Other Mendelian VACTERL genes
42463002	*Surgical management of EA/ TEF in ELBW neonates*	Systematic review	Mortality driven by cardiac/ VACTERL severity
40992427	*Impact of VACTERL and chromosomal anomalies after EA repair (EUPSA)*	Registry	VACTERL high-risk subgroup
35331402	*Long-term digestive outcome of esophageal atresia*	Clinical review	Long-term GI morbidity/QoL
37649433	*Ear anomalies/hearing loss in VACTERL and maternal diabetes*	Clinical cohort	Maternal diabetes risk (OR 3.71)
32924308		Epidemiology	

PMID	Title (abbrev.)	Evidence type	Role in report
	Recurrent constellations of embryonic malformations re-conceptualized		Twinning/ART, early developmental disruption
31519362	*Outcomes in adulthood of gastric transposition for EA*	Clinical cohort	Adult morbidity/QoL data
39998690	*Long-term GI complications in pediatric EA/TEF*	Clinical review	GI surveillance/management

All nine confirmed findings rest on verified abstract quotes. The gene–disease relationship (Findings 1–4, 9) is strongly supported by convergent human genetics, mouse and zebrafish models, population constraint, and ClinVar curation. Prognostic and etiologic findings (6–7) derive largely from the broader VACTERL literature and are applied to WBP11-VCTERL by its membership in that spectrum.

Limitations and Knowledge Gaps

- **Small human cohort.** The founding series is seven families; genotype–phenotype correlation, precise penetrance, and sex ratio for *WBP11* specifically are not robustly quantified.
- **Splicing defect is inferred, not measured in patient tissue.** Step 3 of the causal chain (globally altered pre-mRNA splicing) is biochemically plausible (demonstrated splicing-activating function) but has not been directly demonstrated in affected human organs; which specific mis-spliced transcripts drive each malformation is unknown.
- **SHH-pathway intersection is proposed, not proven** for *WBP11*.
- **Modifiers of expressivity unidentified.** The mechanism branching that yields isolated vs full-spectrum phenotypes is not explained.
- **Prognostic data are extrapolated** from the broader VACTERL/EA literature, not from WBP11-genotyped patients.
- **No WBP11-specific epidemiology, QoL instruments, or natural-history study** exists.

Proposed Follow-up Experiments / Actions

1. **Patient-tissue splicing analysis.** RNA-seq of *WBP11*-haploinsufficient patient-derived cells/organoids (and heterozygous mouse/zebrafish tissues) to identify the mis-spliced developmental transcripts linking the splicing deficit to each organ defect — directly testing inferred step 3.
2. **Mechanistic linkage to SHH/morphogen pathways.** Test whether WBP11 reduction alters splicing/expression of SHH-pathway components (*SHH, GLI, ZIC3, FOXF1, HOXD13, FGF8*) during organogenesis.
3. **Conditional / tissue-specific mouse models.** Organ-restricted *Wbp11* heterozygosity to dissect cardiac vs renal vs esophageal contributions and to model variable expressivity.
4. **Genotype–phenotype and penetrance study.** Aggregate ClinVar/GeneMatcher/registry cases to quantify penetrance, expressivity, and sex ratio for *WBP11* variants, and to test LoF-vs-missense severity differences.
5. **Modifier screens.** Variable-expressivity CRISPR/zebrafish modifier screens to identify genetic/environmental modulators of the branch point.
6. **Clinical surveillance protocol.** Define a WBP11-VCTERL-specific multidisciplinary follow-up pathway (cardiac, renal, esophageal/GI, skeletal) and prospective QoL capture (EQ-5D, GIQLI).

Report compiled from 9 confirmed findings and 29 reviewed papers across 5 iterations. Evidence types: human clinical (defining case series, cohorts, registries), model organism (mouse, zebrafish), in vitro/biochemical (SIPP1 characterization), and computational/population genetics (gnomAD constraint, ClinVar).
